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CHAPTER

7 Perception and Attention

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Abstract

The relation between perception and attention is discussed, via a taxonomy of attention centered around function. Five kinds are distinguished. The first is sampling, involving the pickup of information by the eye. The second is filtering (or gating), the control of information considered relevant. The third is binding, involving the formation of more integrated structure over space. Fourth is holding, which creates the coherent structure necessary to perceive continuity over time. Fifth is indexing, which enables individuation of selected items. The function of each is discussed, along with several associated behavioral effects and perceptual deficits, the mechanisms believed to carry it out, and possible relation to conscious visual experience. Discussion is also given of the implicit processing carried out both prior to and independent of any kind of attention, in the complete absence of conscious visual experience.

Keywords: [attention](#), [awareness](#), [binding](#), [change blindness](#), [consciousness](#), [implicit perception](#), [inattention blindness](#), [selection](#), [taxonomy](#), [tracking](#)

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Our visual experience of the world is one of diverse objects and events, each with particular colors, shapes, and motions. This experience is so coherent, so immediate, and so effortless that it seems to result from a single system that lets us experience everything in our field of view. But however appealing, this belief is mistaken: There are severe limits on what can be visually experienced.

For example, in a display for air-traffic control it is important to track all moving items. For a single item, this can be done without problem. Three or four can also be tracked, although some degree of effort may be needed. As the number is increased further, accurate tracking becomes more and more difficult—and eventually, impossible. Performance is evidently affected by a factor within the observer that enables certain kinds of perception to occur, but is limited in some way. This factor is generally referred to as *attention*.

resource” enabling various operations (see Hatfield, 1998). During the past several decades, considerable progress has been achieved by focusing on the idea of selection (Broadbent, 1982). In particular, attention can be productively viewed as *contingently selective processing*. This can be embodied in various ways by various processes—there need not be a single quantity identified with all forms of attention, or a single site where it operates (Allport, 1993; Tsotsos, 2011). Although “paying attention” is often considered to be a unitary operation, it may simply refer to *the control of one or more selective processes*, ideally in a coordinated way. While this view has some cost in terms of conceptual simplicity, it can help make sense of a large set of phenomena.

p. 98 This chapter surveys several of the major issues in our understanding of attention and how it relates to perception. It focuses on vision, since many—if not all—considerations are similar for all sensory modalities, and the level of understanding achieved in this domain is currently the most advanced. Although many of the issues discussed here also apply to higher level cognitive processes, coverage of those is best done separately.

Kinds of Attention

A great deal of work has been carried out on attention over the years, with a wide variety of approaches used on a wide variety of issues (see, e.g., Parasuraman, 1998; Pashler, 1999; Wright, 1998). Partly because of such heterogeneity, studies of attention have tended to form “silos” that focus on a particular type of experimental approach or proposed mechanism. To provide a more comprehensive view, this discussion includes all of these issues, but it is organized around the possible *functions* that attention might have.

In the view taken here, “attention” is more an adjective than a noun: an attentional process is one that is *contingently selective*, controlled on the basis of global considerations (e.g., selection of a particular item to track, based on perceived importance). Any limited-capacity process is necessarily selective, in that it cannot handle everything; if coupled with appropriate control, it becomes ipso facto attentional. In the case of a limited-resource process, attention is sometimes considered a kind of “fuel” (see Hirst & Kalmar, 1987; Wickens, 1980). However, this does not really help with understanding—it is essentially just a different term for the same problem (Allport, 1989). It may instead be better to discuss processing resources in terms of more concrete measures such as time or processor connections, which if limited can create a need for selectivity, or may constrain a process, but which are not themselves a form of selection or control.

Given all this, it is evident that many kinds of selectivity are possible, each corresponding to a particular kind—or at least, aspect—of attention. These are organized here into a taxonomy describing several of the main kinds possible, as well as the relationships among them. This *function-centered taxonomy* characterizes each kind of attention primarily in terms of its (selective) function. This is followed by the perceptual effects associated with it, the mechanisms that carry it out, and the ways that it enters into our conscious experience of the world.

This taxonomy is incomplete in many ways, and parts of it are provisional. But it encompasses much of what is believed about attention, and provides a natural way to organize a wide variety of results obtained in a wide variety of ways.

Attentional Sampling

Function

The first step of visual perception is the pickup of information by the eye. This is carried out by the selective sampling of incoming light, via a retina with most of its information pickup in the central few degrees of visual angle (see e.g., Barlow, 1981). To compensate for the relatively poor pickup elsewhere, the eye is continually repositioned via brief jumps, or *saccades*. Augmented by head and body movements, these enable access to any part of the incoming light (see, e.g., Henderson & Hollingworth, 1998). A somewhat analogous situation exists in the third dimension (depth), with high-resolution pickup only for the depths that are in focus, a restriction compensated for by continually changing the focus of the lens. When controlled on the basis of global considerations, such *attentional sampling* can enable the right information to be obtained from the environment at the right time, thereby minimizing the computational resources needed for many tasks. In essence, processing over space is traded for processing over time (see Niebur & Koch, 1998). (Also see Chapter 5.)

Attentional sampling—sometimes referred to as *overt attention*—is not the only kind of attention possible. Others also exist, occasionally referred to collectively as *covert attention*. Overt and covert systems are at least partly separate: They need not—and often do not—act on the same information at the same time. For example, although covert spatial attention must go to a location before the eye can saccade there (Deubel & Schneider, 1996), it is not necessary to make—or even intend to make—a saccade to generate a covert shift (Juan, Shorter-Jacobi, & Schall, 2004). Indeed, much of the correlation between these two kinds of attention may arise simply because they tend to respond to the same informative parts of the visual input (Tsotos, 2011, ch. 1).

Perceptual Effects

The limitations of sampling are generally compensated for by appropriate movements of the eye. But they can become evident when such movements are prevented, such as by having the observer fixate a location or view the stimulus during a brief flash. Results show that acuity and color perception are best in the central few degrees (or *foveal area*) and fall off rapidly with increasing eccentricity in the outer parts (or *periphery*); in contrast, motion perception is poor in the fovea but better in the periphery (e.g., Barlow & Mollon, 1982).

Most kinds of eye movement are rapid, and are made several times a second. The particular sequence (or *scan path*) used for viewing an image depends not only on its physical properties but also on the knowledge and interests of the viewer and his or her understanding of the task (Hayhoe & Ballard, 2005; Noton & Stark, 1971).

Mechanism

Many of the systems underlying sampling are fairly well understood. Incoming light is transformed into neural signals by two interleaved arrays of photoreceptors: *cones* (for high levels of illumination) and *rods* (for low levels). Cones are located mostly in the fovea, rods in the periphery. Because cones alone distinguish color, this property is perceived only in the central few degrees of visual angle (see, e.g., Barlow & Mollon, 1982). Meanwhile, an array of *ganglion cells* takes the photoreceptor signals and—after several stages of processing—transmits the results to the brain via the optic nerve. Each ganglion cell receives input from the receptors in a particular area of the retina (its *receptive field*); the size of this increases with eccentricity (see e.g., Kaplan, 1991). Acuity is therefore best in the fovea, where receptive fields are small. Meanwhile, accurate measurement of motion requires the collection of information over a large area, and so motion perception is best in the periphery.

The eye itself is moved via three pairs of muscles, coordinated such that one member of each pair contracts while the other relaxes. These are controlled via several areas in the brain, such as the superior colliculus and the frontal eye fields (see, e.g., Crowne, 1983). And these in turn are under the control of areas involving higher level functions (see, e.g., Carpenter, 1988). (Also see Chapter 5)

Relation to Consciousness

Because information pickup is required for visual perception, attentional sampling is necessary for any conscious (*explicit*) as well as nonconscious (*implicit*) process. But although our visual experience is quite compelling, it does not directly correspond to what is picked up by the retina. For example, the limited range of color and motion sampling is never noticed: We experience the same resolution and color information throughout. Likewise, although saccades are usually made several times each second (often over several degrees of visual angle), our impression is that of a single, stable image. The mechanisms that create such uniformity and stability are largely unknown (Bridgeman, van der Heijden, & Velichkovsky, 1994; O'Regan & Noe, 2001). But their involvement means that although attentional sampling may be necessary for visual experience, it cannot be sufficient.

The involvement of these mechanisms also explains why even though sampling may be necessary for visual experience at a global level, this necessity does not always apply locally. For example, the retina contains an area where the outputs of the ganglion cells exit, rendering it unable to pick up information. But although this *blind spot* may cover as much as 5–7 degrees of angle, it does not enter into our visual experience: Even when the eye is fixated, it appears “filled in,” so the gap is not noticed (Ramachandran, 1992). Similar effects occur for blind spots created by small lesions elsewhere in the visual system, as well as by stimulus manipulations (Ramachandran & Gregory, 1991).

Attentional Filtering

Function

Although a vast amount of information is generally picked up at any moment, most is irrelevant for any particular task. And irrelevant information can be harmful. For example, detection is affected by the noise—or more precisely, the *signal-to-noise ratio*—in the input (e.g., Palmer, Ames, & Lindsey, 1993). Since irrelevant information contains no signal but may contain noise, it can only hurt performance. Likewise, because recognition of a pattern is affected by context (e.g., Oliva & Torralba, 2007), irrelevant background can only interfere. And if not disregarded in some way, irrelevant information can place a load on some processors, causing them to slow down (Tsotsos, 1990). Thus, if perception is to be efficient, it must select the relevant, and only the relevant.

Such *filtering* is in some ways an internal form of sampling. But because it is controlled via changes in neural connections rather than physical forces on the eye, filtering can act more quickly. It can also be more flexible, selecting information from particular regions of space, and with particular ranges of properties (such as color or orientation). This need not be all-or-nothing—a filter can be graded, with inputs weighted as a function of particular values.

p. 100 Two complementary approaches to filtering are possible: selection *for* and selection *against*. Selection *for* relevant information results in the contents being *expressed*. One way of doing so is *gating*—transmitting information at a particular location, or of a particular color, say. In addition, contents can be *enhanced*, having an effect—or at least, priority—above baseline levels. In a similar way, selection *against* irrelevant information results in the selected contents being *suppressed*. When applied to the control of process or behavior, the two approaches are often referred to, respectively, as *facilitation* and *inhibition* (Posner & Snyder, 1975). Both are largely equivalent, although practical considerations may favor one over the other.

Another issue—especially important in the case of space—is the extent of gating. When selection is *diffuse* (input from a large area), considerable information is simultaneously accessible, which can enable several kinds of process to be sped up. However, more noise may be introduced, creating greater risk of error. Conversely, when selection is *focused* (input from only a small area), processing is potentially less error-prone but slower. These are sometimes referred to, respectively, as *diffuse attention* and *focused attention*. There is consequently a trade-off between speed and accuracy depending on the size selected, with the most effective choice depending upon details of the task and environment (see Giordano, McElree, & Carrasco, 2009).

Perceptual Effects

As in the case of sampling, most observers are unaware of the limitations imposed by filtering. But under the right conditions, they can lead to measurable effects. Although not all of these necessarily involve the same neural systems, they are similar in several ways.

Cueing Effects

Observers can detect a target item more quickly and accurately if presented with advance information about its location. This *cueing* occurs even when eye movements are prevented, indicating that it is due to attentional (or covert) *orienting*, the adjustment of internal filters to task demands. These filters block information from irrelevant locations or properties, improving performance by reducing noise.

A transient form of enhancement begins within 50 ms of cue onset and peaks 50–100 ms later. It appears to be due to a *spotlight of attention* that expresses whatever is within the “beam” (Posner, Snyder, & Davidson, 1980) and suppresses whatever is just outside (Steinman, Steinman, & Lehmkuhle, 1995; Tsotsos, 2011). The size of this is about 1 degree at fixation, and it increases with eccentricity (Sagi & Julesz, 1986). It falls off gradually, rather than having a sharp edge (Laberge & Brown, 1989). Its location can be shifted extremely rapidly, at a rate of c. 10 ms/degree (Tsal, 1983) or faster (see Egeth & Yantis, 1997); intermediate locations do not appear to be affected (see Cave & Bichot, 1999). Only one spotlight usually exists at a time (Jonides, 1983), although under some conditions it may be split among several locations (Bichot, Cave, & Pashler, 1999; Kramer & Hahn, 1995; but see Wright & Ward, 2008, ch. 3).

Cueing is also possible for properties such as color and orientation (Humphreys, 1981; Maunsell & Treue, 2006; Rossi & Paradiso, 1995), as well as spatial frequency (Davis, Kramer, & Graham, 1983). It can also affect purely subjective aspects of perception, such as apparent contrast (Anton-Erxleben, Abrams, & Carrasco, 2010; Carrasco, Ling, & Read, 2004).

In general, cueing involves simple properties such as orientation or location. But there is also an influence of larger scale “preattentive” structures—or *segments*—in the background,¹ these apparently formed on the basis of luminance or texture boundaries at early levels of processing, before attentional filtering can act (e.g., Driver, Davis, Russell, Turrato, & Freeman, 2001; Lamme, 1995). For example, cueing is facilitated if cue and target are both in a region interpreted as the surface of the same segment (Egley, Driver, & Rafal, 1994), but not if this region is interpreted as a hole (Albrecht, List, & Robertson, 2008). Consistent with this, cueing appears to operate over perceived rather than retinal space (Robertson & Kim, 1999).

Search Asymmetry

In *visual search*, observers must detect as quickly as possible the presence of a *target* item among a set of *distractors*—for example, a vertical line among a set of obliques. For many kinds of search, the time needed is proportional to the number of items in the display. This has been explained via a spotlight of attention that accesses information from each item in turn, at a rate of 30–50 ms per item (see Wolfe, Klempe, & Dahlen, 2000; but see Thornton & Gilden, 2007). Search slows with increasing eccentricity (Carrasco, Evert, Chang, & Katz, 1995), possibly due to the sparser representation of peripheral areas in the cortex (Carrasco & Frieder, 1997).

p. 101 When the target and distractors differ only by the amount of a simple property (e.g., length or luminance), an interesting *asymmetry* emerges: Search is faster for a longer item among short ones, say, than the other way around. This has been explained by a pooling of the signals in the spotlight. When a target has more of a property, it creates a higher signal. Since accuracy is determined by the ratio of signal to noise, a larger spotlight—which collects more noise—can be used while keeping accuracy the same. Since a larger spotlight requires fewer shifts to cover an area, the result is faster search (Treisman & Gormican, 1988).

Although the properties involved in search may be simple (e.g., length or orientation), they are not directly obtained from simple structures such as line segments. Instead, they appear to be measurements of complex structures—*proto-objects*—of limited extent, formed at early levels prior to attentional filtering (Rensink, 2000a; Rensink & Enns, 1995). Search speed is also affected by background segmentation, being faster for items on the same surface (He & Nakayama, 1995).

Interference Effects

Performance in brief displays can be degraded by nearby items, or flankers (Eriksen & Eriksen, 1974). This *flanker effect* is believed to be caused by a pooling of the information in the spotlight, with signals from the flankers being mixed in with those from the target. Such effects point to a spotlight with a minimum area of 1 degree of angle in the fovea (Eriksen & Eriksen, 1974). It can “zoom out” to a larger area if needed, changing size within 100 ms (Eriksen & St. James, 1986), and automatically becomes smaller when task difficulty or load is increased (Lavie, 1995). Only one such spotlight is believed to operate at a time (Eriksen & Yeh, 1985).

A related effect is *crowding*, where items in the periphery that are easy to recognize when alone become less so when other items are placed nearby; subjectively, their components become “scrambled.” The size of the affected area increases with eccentricity (Bouma, 1970); it can be reduced somewhat if target location is cued (Yeshurun & Rashal, 2010). While part of crowding may be due to poor information pickup in the periphery, much of it appears due to the loss of positional information about the components; indeed, crowding may be much the same as the flanker effect (Strasburger, 2005).

As in the case of cueing and search, interference is influenced by structural factors. For example, it is greater for items grouped by common motion (Driver & Baylis, 1989) or on the same background segment (Kramer & Jacobson, 1991). And the expression of a face can be still perceived even when crowding destroys conscious awareness of the detailed configuration that conveys this (Fischer & Whitney, 2011).

Inattentional Blindness

Consider an observer carrying out a demanding primary task, such as determining which of two similar lines is longest. If an unexpected irrelevant item is shown at the same time, it will often not be seen, especially if less than 1 degree in size. This *inattentional blindness (IB)* has been explained by a failure to “attend” the irrelevant item—that is, a failure to transmit the information from its location (Mack & Rock, 1998). This can last for several seconds, provided that the irrelevant stimulus remains unexpected (Most et al., 2001; Simons & Chabris, 1999). Otherwise, IB virtually disappears, possibly because the filter(s) involved can be reconfigured quickly. IB appears to involve the suppression of items with visual properties similar to those that are ignored (Most et al., 2001).

An important issue is whether the failure to report an irrelevant item results from a failure to *perceive* it or to *remember* that it appeared (Wolfe, 1999). If the primary task is sufficiently demanding, a degree of IB can be induced for repeated stimuli (Cartwright-Finch & Lavie, 2007; Rensink, 2005). Assuming that advance knowledge about the appearance of an item (obtained via the repetition) can prepare the viewer to place a record of it into durable memory at the moment it appears, this indicates that IB is a failure of perception.

Interocular Suppression

In *binocular rivalry (BR)*, two stimuli are simultaneously presented to an observer, one to each eye. But instead of the observer experiencing a combination of the two stimuli, the percept alternates, with only one stimulus experienced at any time (Blake, 2001). This occurs both for simple stimuli and complex events (Neisser & Becklen, 1975), and it likely results from the suppression of information on the basis of eye of origin.

A related effect is *continuous flash suppression (CFS)*, where a set of random images flashed into one eye at a rate of about 10 Hz completely suppresses the experience of an image shown to the other (Tsuchiya & Koch, 2005). This effect can be sustained for several minutes. The degree of suppression is more severe than that found in BR, although some of the same neural systems may be involved (Tsuchiya, Koch, Gilroy, &

Blake, 2006). The afterimage of a suppressed stimulus is weakened, consistent with a suppression of information even to mechanisms not directly involved with conscious perception (Kaunitz, Fracasso, & Melcher, 2011; Tsuchiya & Koch, 2005).

Neglect (Hemineglect; Unilateral Neglect)

Damage to the right posterior parietal cortex can result in *neglect*—the absence of visual experience in some part of space (Bartolomeo & Chokron, 2002; Bisiach, 1993). Two variants can be distinguished. In *egocentric* neglect, the absence is of one side—usually the left—of viewer-centered space (Posner, Walker, Friedrich, & Rafal, 1984); in *allocentric* neglect, absence is of one side—usually the left—of each item in the entire visual field (Driver & Halligan, 1991). A related condition is *extinction*, where an item vanishes when a competing item appears on the opposite side of the visual field. These deficits appear to result from a failure to attentionally gate—that is, transmit—information from the appropriate area (or structure), causing either an outright failure to access the information it contains or at least a slowdown in the speed of processing (Posner et al., 1984). These deficits can be influenced by the structure of unseen objects, as well as by background segmentation (Rafal, 1994).

Mechanism

The effects described earlier appear to involve a mechanism with a characteristic mode of operation: It acts within a very brief time (c. 50 ms), involves simple variables (such as spatial location, orientation, and color), and for spatial variables operates like a reconfigurable spotlight. It is modulated by a retinotopic (i.e., eye-centered) organization, with performance deteriorating as eccentricity increases, likely due to the sparser representation of peripheral areas in the cortex. The filtering common to all of these is sometimes referred to as *spatial attention* or *space-based attention* (see Cave & Bichot, 1999). The information transmitted does not seem to keep track of its precise position in the visual field; when it is pooled, confusions often arise.

Although the variables that control filtering are simple, the structures it operates upon are not, having at least some degree of organization (also see Duncan, 1984; Egeth & Yantis, 1997). Background structures—*segments*—appear to facilitate information access (also see Franconeri, Bemis, & Alvarez, 2009; Kimchi, 2009). Filtering, in turn, can influence these, possibly via feedback connections (Driver et al., 2001). Likewise, although the variables transmitted (such as length) are simple, they are derived from relatively complex proto-objects in which various properties have already been bound together to some extent, presumably at low levels (Rensink, 2000a; Rensink & Enns, 1995). Also see Perceptual Organization in Vision (Chapter 2).

Models of these mechanisms include those of Posner et al. (1980) and Treisman and Gelade (1980). Neural implementation may involve the dynamic restriction of receptive fields of neurons in visual cortex (Desimone & Duncan, 1995; Moran & Desimone, 1985). Control has been modeled by a system with three components: disengagement, shifting, and reengagement (Posner et al., 1984). A more general framework for filtering is that of Nakayama (1990). Detailed computational models include those of Cave (1999), Niebur and Koch (1998), and Tsostos (2011). (Also see Chapter 8.)

Relation to Consciousness

Much of consciousness appears to involve the consolidation of information over a global scale (Baars, 2005; Cohen & Dennett, 2011). This potentially connects to attention in at least two ways. First, conscious experience might be involved in transmitting the *output* of an attentional process elsewhere. Second, given that an attentional process can be responsive to global considerations, experience may enter into the control of its *operation*—for example, the override of automatic settings (Libet, 1985). As such, there may well be an association between conscious experience and at least some aspects of attentional filtering.

Conscious Perception

Attentional filtering—or at least, gating—appears necessary for conscious visual experience. In binocular rivalry, for example, observers simply fail to experience anything from the unmonitored eye (see Rensink, 2009). A similar failure occurs in inattention blindness when the irrelevant stimulus is less than a few degrees (Mack & Rock, 1998), and in neglect, where gating has presumably failed entirely (Posner et al., 1984).

p. 103 An even deeper relation may exist. When given a primary task of identifying a pattern at the center of a brief display, observers can still detect simple *features* elsewhere, such as localized colors or orientations (Braun & Sagi, 1990). They can also rapidly determine the meaning—or *gist*—of an image (Li, VanRullen, Koch, & Perona, 2002; Oliva & Torralba, 2006) and summary statistics such as average size of items (Ariely, 2001; Chong & Treisman, 2003). Here, gating may occur over a relatively large expanse, possibly in a sequential way (VanRullen, Reddy, & Koch, 2004). As the primary task becomes more demanding, gist perception begins to fail (Cohen, Alvarez, & Nakayama, 2011). If this is due to reduced gating (Lavie, 1995), it would suggest that gating—but possibly no other kind of attention—is necessary for any process concerned with the fragmentary aspects of an image.

The perception of these fragmentary aspects appears to correlate with a distinct kind of subjective experience. When an irrelevant IB stimulus is sufficiently large and the primary task not too demanding, observers often report a “fragmented” experience, being able to see “something” but not being able to say what it is (Neisser & Becklen, 1975). Such a base-level—or *ambient*—experience may be similar to the *background consciousness* of briefly presented displays (Iwasaki, 1993), where contents are fleeting, fragmentary, and seem to contain more than can be reported. It may also be related to the *phenomenal- or P-consciousness* taken to describe raw experience (Block, 1995; also see Lamme, 2003).

Nonconscious Perception

Visual processing in the complete absence of visual experience—and presumably, spatial gating—is evident in many situations. During binocular rivalry, for example, faces and familiar stimuli in the unmonitored eye are not experienced, but they can enter conscious awareness more quickly (Jiang, Costello, & He, 2007). And during inattention blindness, an unseen item can cause *priming*—speeding up the identification of a related item that is presented soon afterward (Mack & Rock, 1998). It can even draw attention to itself (Mack & Rock, 1998; but see Harris & Pashler, 2004).

Such effects are likely related to those found in *subliminal perception*, where observers generally fail to consciously experience a target that is presented extremely briefly (c. 20 ms), presumably because attentional processes are not given sufficient time to operate. But although not consciously experienced, stimuli presented under these conditions can still cause priming; interestingly, this occurs only when the time of their appearance is predictable, indicating the involvement of (possibly automatic) temporal gating (Naccache, Blandin, & Dehaene, 2002).

More generally, effects such as these suggest the existence of what might be called “*dark structure*” — structure that never can be part of visual experience, yet still affects perception. Whether the structures that influence filtering are also dark, or just the initial stages of structures that are later experienced, is an open question.

Attentional Binding

Function

At some point, the pieces of information obtained in the initial stages of vision must be *integrated*. This already occurs to an extent at early stages of processing, for example, the creation of background segments and proto-objects (see section on “Mechanism”). But to better capture the structure of the world, a more sophisticated structure is needed, one in which *all* the properties relevant at a moment in time are appropriately linked and are connected to a particular position in space.

One aspect of this is the linking of properties at the same location. This can occur at early levels of visual processing, with some neurons responding to particular combinations of simple properties (see e.g., Sincich & Horton, 2005); it also shows up at higher levels (e.g., van Dam & Hommel, 2010). But other aspects also exist, which may involve greater distances, greater precision, greater numbers of properties—and therefore, greater challenges (see Tsotsos, 2011, ch. 6). And all of these may require that structures be created—or at least modified—on the basis of global considerations. In other words, integration may require *attentional binding*.

Attentional binding is often associated with attentional filtering. Indeed, the two are sometimes considered to be much the same, with binding referred to as “focused attention.” But binding involves rather different functions. For example, if a line segment cannot be assigned to a particular group on the basis of purely local factors, its assignment must be based on more global considerations. Similar concerns exist for other aspects of structure, for example, high-resolution estimates of orientation. More generally, the determination of integrated structure requires solving the *binding problem*: how to correctly link all relevant components (Robertson, 2003; Treisman, 1999). As such, binding is considered here to be a separate attentional process.

Perceptual Effects

p. 104 The limitations of attentional binding show up in various ways. As in the case of filtering, not all ↴ necessarily involve the same neural systems. But they do appear to have several elements in common.

Conjunction Search

Experiments on visual search (see section on “Neglect”, although I’ll leave that up to you) show that the time needed to detect a unique combination (or *conjunction*) of simple features is proportional to the number of items present. This is believed to be due to a spotlight of attention that binds the features at each location, at a rate of 30–50 ms per item (Treisman & Gelade, 1980; Wolfe, 1998; but see Eckstein, 1998). Search is slower at greater eccentricity, perhaps due to sparser representation in the corresponding cortical areas (Carrasco et al., 1995; Carrasco & Frieder, 1997). Comparable rates are found when spatial form needs to be resolved (Duncan & Humphreys, 1989; Kleiss & Lane, 1986).

Not all combinations need attentional binding. Relatively fast search is possible for some scene properties and groups (Rensink & Enns, 1995, 1998), suggesting that it can access proto-objects that already contain a

considerable degree of binding (but see VanRullen et al., 2004). In addition, some kinds of conjunction search can be sped up if the target is known ahead of time and the items are relatively far apart. Such *guided search* may occur because items with one of the target properties are picked out via filtering, resulting in a simpler task involving the remainder (Treisman & Sato, 1990; Wolfe, 1994). If the selection of one property in a structure can result in the immediate selection of the others, a possible explanation is that some degree of binding is in place prior to attentional processing.

Illusory Conjunctions

When nearby items with different properties are displayed briefly (e.g., 200 ms), viewers often report seeing the properties combined incorrectly (Prinzmetal, 1995; Treisman & Schmidt, 1982). These *illusory conjunctions* have been explained in terms of a process that requires 100–200 ms to link the features within the spotlight of attention and determine a definite position for each. If insufficient time is given, positional information remains poorly established, creating the possibility of erroneous bindings (Cohen & Ivry, 1989).

In foveal vision, illusory conjunctions occur primarily when items are within 1 degree of each other (Cohen & Ivry, 1989). In peripheral vision, they can occur for larger separations and slower speeds (Prinzmetal, Henderson, & Ivry, 1995), possibly because crowding (see section on “Neglect”, although I’ll leave that up to you) increases the difficulty of access to individual features (also see Pelli, Cavanagh, Desimone, Tjan, & Treisman, 2007). Illusory conjunctions occur more frequently within perceptual groups (Prinzmetal, 1995), showing the influence of background structures (such as segments) that contain a degree of organization. They are unlikely if the binding would create a semantic association not in the original items—for example, there is little illusory conjunction of circle plus black if the result would be a tire not present originally (Treisman, 1986).

Repetition Blindness

If an observer looks for a target item in a stream of briefly presented stimuli (100–200 ms each), a repeated target is much less likely to be seen than a different item at the same position in the sequence. Such *repetition blindness (RB)* is thought to occur because the second item is perceived as an instance of a category (or type) encountered recently, which prevents triggering the formation of a new integrated structure (Kanwisher, 1987).

The type involved in RB is somewhat abstract, being invariant to size, orientation, and viewpoint (Kanwisher, Yin, & Wojciulik, 1999). This suggests it can be determined within about 100 ms, which in turn suggests that the process involved draws upon structures that already have some degree of binding. Only those items that are filtered in (e.g., those with a particular color or orientation) appear to be relevant for determining what constitutes a repetition (Kanwisher, Driver, & Machado, 1995).

Integrative Agnosia (Apperceptive Agnosia; Visual Space Agnosia)

Damage to the inferior temporal lobe sometimes results in *integrative agnosia*, an inability to perceive overall shape or configuration, with perception being only of simple features, such as color or texture (Farah, 2004; Riddoch & Humphreys, 1987). This is likely caused—at least in part—by a disabling of attentional binding. The ability to determine which items belong together appears to be unaffected, consistent with the idea that several aspects of (nonattentional) binding are carried out prior to attentional binding (Behrmann & Kimchi, 2003).

Mechanism

p. 105 The effects described earlier involve bound structures that enable processes such as recognition of complex shapes and characters. Most indicate a mechanism with a time constant of about 150 ms—shorter than the 300 ms typical of a fixation, but longer than the 50 ms characteristic of filtering. The 50 ms value encountered in conjunction search may reflect a mechanism based on filtering (Treisman & Gelade, 1980), but it could also simply reflect the time needed to gate an item into a pipeline that operates on several items concurrently, with each processed for several hundred milliseconds (Wolfe et al., 2000).

The mechanism involved appears to operate upon segments and proto-objects that already have a degree of binding, much—if not exactly—the same as the structures operated upon by filtering. Indeed, *feature integration theory* (Treisman & Gelade, 1980) posits that filtering is the basis of a good part of attentional binding. For example, when searching for a conjunction of different kinds of properties, or *dimensions* (e.g., a red T among red As and blue Ts), it proposes that the spotlight of attention restricts gating to a small location, resulting in the simultaneous detection of all features there.

But for many tasks, attentional filtering and attentional binding appear to be carried out by different mechanisms (Bravo & Nakayama, 1992; Briand & Klein, 1986; Maddox, Ashby, & Waldron, 2002; VanRullen et al., 2004); these may involve, respectively, the *transient* and *sustained* forms of attention that are sometimes distinguished (Nakayama & Mackeben, 1989). In addition, the resolving of fine detail appears to involve still another mechanism, one that combines information from filters operating within a single dimension (Laberge, 1998).

The neural systems that implement (attentional) binding are largely unknown (Robertson, 2003). One possibility is the synchrony in firings of neurons that belong to the same group (Singer, 1999). Another—not necessarily exclusive—account is based on the dynamic, selective routing of information from various areas of cortex (Olshausen, Anderson, & Van Essen, 1993), likely involving feedback connections between several levels of processing (Di Lollo, Enns, & Rensink, 2000; Lamme, 2003; Tsotsos, 2011, ch. 7).

Relation to Consciousness

Just as conscious experience is associated with some aspects of attentional filtering, it may also be associated with some aspects of attentional binding. And as in the case of filtering, this could be in terms of both output and operation.

Conscious Perception

Attentional binding appears necessary to experience structure containing integrated properties, spatial extension, and a well-defined position. This kind (or perhaps level) of *assembled experience* comprises ambient experience (providing basic sensory properties) along with a degree of additional (bound) structure; it may be related to the *access- or A-consciousness* involved with verbal reports (Block, 1995; also see Lamme, 2003). If attentional binding is diverted to some task, irrelevant visual stimuli can still be experienced, but only in ambient form, either the fragmentary features themselves (Braun & Sagi, 1990) or properties based directly upon them, such as scene gist (Li et al., 2002; VanRullen et al., 2004). Consistent with this, ambient experience alone appears to be encountered in integrative agnosia, likely due to the loss of attentional binding.

There is some possibility that attentional binding is involved in creating the “figure” in figure-ground organization (Vecera, Flavaris, & Filapek, 2004), while other kinds of attention—such as those involved in change detection—are not (Kimchi & Peterson, 2008). Given that “figure” is generally experienced in an

assembled way (and “ground” in an ambient one), it may be that attentional binding is associated at least to some extent with assembled experience.

Nonconscious Perception

In integrative agnosia, the overall configuration of an object is not experienced in an assembled way (i.e., as a bound structure) but is still available for motor tasks such as grasping (Goodale & Milner, 1992). This suggests that integrated structure is still captured in other processing streams, possibly as dark structure that is never experienced consciously. Indeed, even when background organization is not seen in an assembled way, it can still affect the perception of stimuli that are (Driver et al., 2001; Moore & Egeth, 1997). Such phenomena are consistent with the proposal that attentional binding is involved in the control of processes that have already captured a considerable degree of structure.

Attentional Holding

Function

p. 106

Although binding is important for various tasks, it is not always enough. For example, when a physical object changes, it is helpful to see it as a single *visual object* that endures over time, rather than a sequence of unrelated structures. Such *continuity* likely plays a central role in the perception of physical objects and events. However, establishing this additional level of structure is a complex process, in that both permanence (persistence of identity) and change (transformation of properties) must be handled simultaneously (Kahneman, Treisman, & Gibbs, 1992). Also see Object Recognition (Chapter 3).

Such continuity can be achieved via the idea of *coherent* representation. Here, properties are linked not only across space but also across time, so as to refer to a single persisting object (Rensink, 2000a). As in the case of binding (and to some degree, filtering), this process may take time. Indeed, owing to the greater complexity of coherent structure, such *attentional holding* may require considerably more in the way of various computational resources.

Perceptual Effects

Similar to the situation for other kinds of attention, the limitations of attentional holding show up in various ways, with several common characteristics.

Object-Specific Preview Benefit

When several items in a display are presented briefly (250 ms), a letter or digit located inside one of them can be identified more quickly the next time it appears, provided it remains inside the original item. This *object-specific preview benefit* lasts for over 600 ms and can survive considerable displacements of item location (Kahneman et al., 1992) and even the addition of a border that does not overlap existing components (Gao & Scholl, 2010).

This effect weakens as the number of items increases from 2 to 4, suggesting that some aspects of the process involved are extremely limited, possibly to one item at a time. The formation of the representations underlying this effect is believed to be largely complete by about 250 ms (Kahneman et al., 1992).

Change Detection/Change Blindness

If a change is made while the accompanying motion signals are swamped (e.g., by an image flicker or eye movement), it can be quite difficult to notice, even when large and continually repeated (Rensink, O'Regan, & Clark, 1997; Simons & Levin, 1997). This *change blindness* is believed to reflect a failure to attentionally hold the item at the moment of change (Rensink et al., 1997; Rensink, 2000a). Although reminiscent of inattention blindness (see section on "Neglect", although I'll leave that up to you), the two are not the same; for example, change blindness can still occur when the observer is aware that a change could happen, suggesting that the underlying processes are less flexible, or at least, slower (see Rensink, 2009).

When detecting the presence of change, up to 3–5 items can be monitored at a time (Luck & Vogel, 1997; Pashler, 1988; Rensink, 2000b). Relatively little information about each is stored (Alvarez & Cavanagh, 2004), and not all structural levels are represented (Austen & Enns, 2000). Filtering is able to control the properties considered relevant (Rensink, 2000b). When detecting an *absence* or *conjunction* of changing properties, only one item can be compared at a time, at a rate of about 300 ms per item. Thus suggests that information from the monitored items is not maintained in a set of independent representations but is collected into a single *nexus* (see Rensink, 2002).

Third-Order Motion

Three different kinds of motion are often distinguished. *First-order* motion is supported by differences in luminance and results from the operation of low-level detectors (Cavanagh & Mather, 1989; but see Petersik & Rice, 2006). *Second-order* is encountered for patterns with no luminance differences (e.g., textures or isoluminant colors); it appears to be based at least in part on the displacement of low-level structures (Seiffert & Cavanagh, 1998). Finally, *third-order* motion involves the tracking of movement patterns via higher level operations. It acts on extended spatial structures (e.g., a group of lines), and it can survive changes in their underlying properties (Lu & Sperling, 2001).

The perception of third-order motion is believed to require at least one kind of attention (Cavanagh, 1992). This differs from the kind that enables character recognition (presumably binding), although it may draw upon it. The mechanism involved has a time constant of about 300 ms; only one movement pattern can be seen at a time (Ho, 1998).

Attentional Blink

If two prespecified targets in a stream of briefly presented stimuli appear at slightly different times, an observer will often see the first target but not the second. This phenomenon is called the *attentional blink* (AB); it is thought to result from a failure to attentionally hold the second target, possibly because the representation of the first is not yet complete (Raymond, Shapiro, & Arnell, 1992). The mechanisms underlying this effect are selective. They differ from those that underlie repetition blindness but may involve coherent representation of some kind (Chun, 1997). The meaning of a word not consciously experienced during the blink can be implicitly perceived (Luck, Vogel, & Shapiro, 1996).

The separation needed for AB is 250–300 ms, a value known as the *attentional dwell time* (Duncan, Ward, & Shapiro, 1994; Theeuwes, Godijn, & Pratt, 2004). Only one such representation appears to be created at a time.

Simultanagnosia

Simultanagnosia is an inability to recognize more than one object at a time, resulting from damage to the brain (Coslett & Saffran, 1991). Two variants are known. The first, *dorsal* simultanagnosia, results from bilateral damage to the junction between the parietal and occipital lobes. It involves a failure to see more than one object or part of an object, with the rest of the input simply not experienced (Farah, 2004). The meaning of unseen items, however, can still have an effect (Jackson, Shepherd, Mueller, Husain, & Jackson, 2006). The second variant, *ventral* simultanagnosia, is caused by damage to the left inferior temporo-occipital region. It also involves a failure to recognize more than one object, but allows perception—including counting—of several simple shapes simultaneously (Farah, 2004). Both variants are believed to stem from an impairment of attentional processing (Coslett & Saffran, 1991; Duncan et al., 2003), likely including the engagement or disengagement of attentional hold, and—for dorsal simultanagnosia—attentional gating as well.

Mechanism

The effects described earlier involve a mechanism—or set of mechanisms—with many of the following characteristics: relatively slow (c. 300 ms) access to only a few structures and a small amount of information from each, ability to filter inputs via simple properties, and only one overall structure in play at a time. Such a mechanism could represent at most only a few aspects of a physical object at any time. But if the relevant quantities could be created when requested, the result would be a *virtual representation* of the object, with any relevant aspect represented whenever needed (Ballard, Hayhoe, Pook, & Rao, 1997; Rensink, 2000a).

One model of this is the *object file*, a temporary representation of (bound) properties intended to capture the continuity of an object as it transforms or changes position (Kahneman et al., 1992). Containing only a handful of features and a single level of hierarchical structure, it forms a highly compact description of the object (including the spatial relations among its parts). Only a few object files are assumed to exist simultaneously, with only one updated at a time.

A proposal similar in many ways is the *coherence field*, in which information is “held” in reverberating circuits implemented via feedforward and feedback connections between selected proto-objects and a higher level nexus, with only a few properties in play at any moment (Rensink, 2000a). The selected proto-objects become part of a coherent visual object; when the circuit is broken, the object-level structure simply dissipates, without any accumulation of information in coherent form (also see Wolfe et al., 2000). Some information can be retained—although no longer in coherent form—for a small number of items (perhaps 10 or so), possibly as a memory to help guide subsequent allocation (Hollingworth & Henderson, 2002; Rensink, 2000a).

A more detailed quantitative model that handles many of the issues involved with the representation of coherent (and bound) structure is the *Visual Theory of Attention* (Bundesen, 1990). This incorporates several aspects of attentional processing in a unified way and provides a good fit to a considerable body of experimental data (e.g., Duncan et al., 2003).

In all these proposals a critical role is played by visual short-term memory (vSTM), which provides continuity over durations greater than a few hundred milliseconds. Although the contents of vSTM play an important role in coherent representation, the two are not necessarily the same, at least conceptually; the contents of vSTM, for example, can be maintained for several seconds after visual input disappears, whereas coherence collapses within a few hundred milliseconds afterward (Rensink, 2000a). Bindings between different dimensions appear to be maintained in vSTM (Gajewski & Brockmole, 2006; but see Wheeler & Treisman, 2002), whereas the status of these in coherent representation (where they are held only over

short durations) is unknown. And while more sophisticated structures, such as movement patterns—or *sprites*—may be accessed by vSTM to enable the recognition and anticipation of complex movements (Cavanagh, Labianca, & Thornton, 2001), it is likewise unclear whether these are also part of a coherent representation. More generally, the heavy involvement—if not outright embedding—of vSTM in long-term memory (Cowan, 2001) might enable access to considerable knowledge about possible movements and other transformations, which could *guide* the creation of coherent representation. ↵ But such knowledge need not be contained in the representation itself.

Relation to Consciousness

As in the case of other kinds of attention, conscious experience is likely involved with at least some aspects of attentional holding, in regard to both output and control of operation. Indeed, the likelihood of such involvement is highest for this kind of attention. Given that coherence is expensive in terms of computational resources, it is extremely important to guide its usage effectively, both when transmitting it to the appropriate processes and deciding when to create it in the first place. The more global the basis of this guidance, the better.

Conscious Perception

Attentional holding appears necessary for a kind (or possibly level) of *coherent experience*, which includes not only ambient (and perhaps assembled) experience but also the impression of a persisting substrate. This may be similar to the *object consciousness* proposed on the basis of verbal reports of structure, which is believed to involve vSTM (Iwasaki, 1993). In the absence of attentional hold, coherence does not exist; experience is of a succession of static structures, somewhat akin to perception under stroboscopic conditions.

Whether attentional hold is also sufficient to perceive a coherent object depends on what is meant by “object.” If this refers to a physical object, hold is clearly insufficient: Relatively little information is maintained in coherent form, and so most properties are not experienced in a coherent way. But if the term refers to a visual object—that is, the contents of a coherent representation—it may be both necessary *and* sufficient (Rensink, 2002).

Two types of change can be distinguished on the basis of subjective impression (Rensink, 2002). *Dynamic* change is experienced as the transformation of a structure that is *continually present* throughout the process. In contrast, *completed* change is experienced as an event that has simply *happened* at some time in the past. These appear to involve different time constants. Dynamic change cannot survive an interstimulus interval much beyond 300 ms, while completed change is far more robust; it also appears to be the only type experienced for changes made during eye movements (Rensink, 2002; also see Hollingworth, 2008). It may be that dynamic change involves coherent representation, and completed change, the contents of vSTM alone.

Nonconscious Perception

Although attentional hold may be necessary for the visual experience of change, implicit detection of change may still occur without it (Fernandez-Duque & Thornton, 2000; Laloyaux, Destrebecqz, & Cleeremans, 2006). The existence of this is controversial (Mitroff, Simons, & Franconeri, 2002). But if it does exist, it may reflect the operation of low-level feedback circuits that provide a degree of continuity over brief temporal gaps (cf. Di Lollo et al., 2000).

Another interesting phenomenon is the report of some observers of “sensing” or “feeling” a change without an accompanying visual experience. This may be a form of nonvisual—but still conscious—awareness involving mechanisms that differ to some extent from those underlying visual experience (Rensink, 2004). Such an account is controversial (Simons, Navarez, & Boot, 2005), but additional support for it has been found (Busch, Fründ, & Herrmann, 2010; Galpin, Underwood, & Chapman, 2008).

Attentional Indexing

Function

It is sometimes important to individuate a physical object—to see it not just as *an* object (coherent or otherwise) but as a *particular* object. This is especially so when more than one object must be dealt with, for example, determining spatial relations such as “between,” assessing the configuration of a group, or ensuring that items in an image are processed in an optimal—or at least, effective—sequence (Pylyshyn, 2003; Ullman, 1984).

Although this might be done via multiple coherent representations, it is difficult to create more than one such representation at any time (see section on “Mechanism”). But fortunately, simultaneous coherence is rarely needed. Many operations involve only a few items, with each needing only to be distinguished in some way, such as by location. Consequently, an item only needs to contain enough information to be considered distinct at some point, and then kept track of as it moves about. If such *indexing* is successful, the item can be immediately accessed whenever needed.

Perceptual Effects

Although indexing can be useful, it also has limitations. As in the case of other kinds of attention, these need not all involve the same neural systems. But they nevertheless appear to exhibit a degree of similarity.

p. 109 Multiple-Object Tracking

In *multiple-object tracking (MOT)*, a set of identical items is displayed, a subset marked (e.g., briefly flashed), and the marked items tracked for several seconds as all items move around (Pylyshyn & Storm, 1988; Scholl, 2009). Accuracy is often good up to about 4 tracked items, this limit varying between 1 and 8, depending on task demands (Alvarez & Franconeri, 2007). Accuracy also depends on eccentricity. It is limited by a resolution coarser than that due to purely sensory limitations, possibly reflecting the involvement of attentional gating (Intriligator & Cavanagh, 2001). Indeed, tracking may help with the control of gating (Bettencourt & Somers, 2009). But it does not improve change detection (Bahrami, 2003), suggesting that it cannot help control holding.

Although tracked items are often considered independent, they can be perceived as the corners of a virtual polygon that deforms over time (Yantis, 1992). Tracking is not extrapolated when an item becomes occluded (Franconeri, Pylyshyn, & Scholl, 2006), and it does not appear to facilitate binding of the properties of the tracked item or their encoding into vSTM (Bahrami, 2003; Scholl, 2009). Meanwhile, it is possible to track the center of a simple extended structure (Scholl, Pylyshyn, & Feldman, 2001), but not the intersection perceived to result from two moving structures (Anstis, 1990). This suggests that tracking is of structures rather than simple features, but that these structures are not highly integrated, perhaps being preattentive segments or proto-objects. Tracking is based on perceived rather than retinotopic space (Liu et al., 2005).

Prioritization of Search

A target item in visual search (see section on “Neglect”, although I’ll leave that up to you) is more rapidly detected if in a set of items whose locations are shown ahead of time. This *prioritization of search* is believed to reflect the existence of a set of preferred locations (Yantis & Jones, 1991; but see Watson & Humphreys, 1997), each corresponding to an indexed item.

Up to four locations can be prioritized this way; the effect does not seem to be affected by the distance between them (Burkell & Pylyshyn, 1997; Yantis & Jones, 1991). Considerably more can be prioritized under some conditions, but this may simply reflect a group that is perceived as a single indexed item (Belopolsky, Theeuwes, & Kramer, 2005).

Inhibition of Return

When an item is attentionally gated, more time is needed to respond to it for a short while afterward (Posner & Cohen, 1984). This *inhibition of return (IOR)* takes at least 200–300 ms to develop (depending somewhat on task demands), and it can last several seconds. It appears only after the application of attentional operations involving space (Klein, 2000). It is believed to mark particular items or locations, so as to prevent attentional processes from returning too soon to a previously visited input (see Klein, 2000).

At least four locations (items) can be inhibited simultaneously; distance of separation does not appear to matter (Snyder & Kingstone, 2001; Wright & Richard, 1996). IOR acts in an object-centered rather than a retinotopic frame (Tipper, Driver, & Weaver, 1991).

Subitizing

Subitizing is the rapid, effortless, and accurate counting of a small number of items. Up to four can be enumerated this way, at a rate of about 50 ms per item; beyond this, a slower process is needed (Trick & Pylyshyn, 1994; but see Piazza, Mechelli, Butterworth, & Price, 2002). Individuation is clearly involved, so as to ensure that the subitizing is of distinct elements.

Arbitrary combinations of features cannot be subitized, although localized groups can (Trick & Enns, 1997), suggesting that the process acts on structures (e.g., proto-objects) formed prior to attentional binding (Pylyshyn, 2003; Scholl, 2009). There appears to be no effect of spatial separation, but performance is slower for items on the same segment (or bound structure), indicating that these items are not considered separate, at least initially (Davis & Holmes, 2005).

Mechanism

Each of the effects described earlier appears to involve a mechanism that can index about four items. Position is referenced not to an arbitrary point in retinotopic space, but to a structure—a segment, proto-object, or frame—created at early levels of processing. Selectivity can be allocated on the basis of global considerations, showing that this process is attentional (also see Scholl, 2009). But it operates rapidly and appears to be concerned with facilitating the control of other processes, perhaps explaining why it has not always been considered a form of attention (cf. Pylyshyn, 2003).

p. 110 Coherent representations do not appear to be automatically indexed (van Dam & Hommel, 2010), suggesting that indexing may be carried out by mechanisms largely separate from those used for holding (also see Bahrami, 2003; Scholl, 2009). One proposed mechanism is the *fingers of instantiation (FINST)* model, which posits that 4–6 items can be simultaneously tracked (Pylyshyn, 2003; Pylyshyn & Storm, 1988). A variant is the *flexibly allocated index (FLEX)* model, in which 1–8 items can be tracked, with a minimal separation

between them (Alvarez & Franconeri, 2007). The memory involved in indexing may be related to a spatial short-term memory concerned exclusively with location (e.g., Darling, Della Sala, & Logie, 2007; Xu & Chun, 2006).

Relation to Consciousness

At the subjective level, indexing requires effort, consistent with it being a form of attention (see Scholl, 2009). It also appears to involve—at least during tracking—an experience of continuity, one similar to that experienced during holding, where objects maintain coherence over both time and space.

Whether indexing is also associated with a distinct kind of perceptual experience (corresponding to individuation, say) is an open issue. Given that it largely acts by facilitating the control of other attentional processes, the kinds of experience associated with indexing may not extend beyond those associated with other kinds of attention. More precisely, although indexing may enable more effective control of attentional processing, its effects may be exhibited mostly—if not entirely—via the facilitation of mechanisms found elsewhere; no new kinds of output or operational control may be involved. As such, no new kinds of conscious experience would be expected. The definitive resolution of this issue will likely require additional work in which the effects of indexing can be separated out from those of the processes it facilitates.

Conclusion

Although the relation between perception and attention is complex, it is possible to clarify—and potentially, simplify—several of the issues involved. For visual perception, a *function-centered taxonomy* can distinguish at least five kinds of attention, each having a distinct function and characteristic mode of operation. Many of the behavioral effects associated with attention (including perceptual deficits) can be identified with one of these. In addition, at least three kinds of visual experience can be distinguished, involving increasingly sophisticated levels of represented structure, and each associated with a particular kind of attention. The results of many studies also indicate that processes of considerable sophistication occur both prior to and independent of any kind of attention, in the complete absence of conscious visual experience.

Future Directions

1. *What exactly is an attentional kind?* The taxonomy used here centers around function, with each function appearing to be associated with a distinct set of effects and mechanisms.² But how real is this? Does each kind of attention correspond to an aspect or stage of a unitary process, or to something more? To answer this would include determining whether all effects believed to reflect the same kind of attention actually involve the same mechanism; this in turn would require careful measurement of things such as timing parameters and variables involved. It would also include determining the extent to which each mechanism is also used for other functions (e.g., a mechanism for filtering that is also used for binding). Some of these issues could be investigated via approaches such as the *attention operating characteristic* (AOC), which shows the degree to which the same mechanism (or “resource”) is used in two concurrent tasks (Sperling & Doshier, 1986).
2. *What kinds of attention exist?* Over the years, various processes have been lumped together under the name of “attention” (Allport, 1993; Chun, Golumb, & Turk-Browne, 2011; Tsotsos, 2011). It is important to determine what these are in a rigorous way. The approach used here might be extended to other kinds of contingently selective process, for example, assignment of semantic associations or control of motor actions. Additional kinds of attention might also be found via the splitting of existing categories—for example, binding separated into the linking of features and the formation of extended configurations across space. Progress here would include developing more rigorous criteria for determining what counts as a distinct kind of attentional process, as well as a more careful examination of effects currently considered to involve the same kind of attention.
3. *How do the various kinds of attention relate to each other?* This is essentially the generalization of the question of how eye movements relate to covert kinds of attention. A key step is to determine which
↳ attentional processes depend on which. Although there could in principle be interactions among several processes simultaneously, much could likely be done simply by investigating pairs of interactions. If there is some dependence between particular processes, it becomes important to determine how closely their operation is aligned, both in space (operands selected) and in time (kinds of delays).
4. *What are the characteristics of the processes operating prior to attention?* The kinds of attention discussed here all appear to operate on the same preattentive structures: extended segments, localized proto-objects, and possibly frames of some kind. It is important to get a better understanding of these. Do they form a common substrate for all attentional processes? Are they distinct or different aspects of the same thing? To what extent can they be overridden—or at least, set—by attentional control allocated ahead of time? These questions can be investigated via modifications of the techniques outlined here.
5. *How do the different kinds of visual experience relate to the different kinds of attention?* Interest in how attention relates to conscious awareness spans centuries. Most proposals (e.g., Koch & Tsuchiya, 2007; Lamme, 2003) assume only one kind of attention and one kind of visual experience. However, given the distinctions that appear to be emerging, this issue needs to be recast into a more articulated form: Relationships may only exist between some kinds of attention and some kinds of experience. Given the number of possible combinations and the number of issues to consider in each, establishing such relationships will take considerable work. But once determined, they will likely give us important new insights into the nature of both attention and consciousness.

Acknowledgments

This chapter is dedicated to the memory of Jon Driver.

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Notes

1. These have often been referred to as “objects.” But this word is overworked, having been used to refer to virtually any level of structure. Following Driver et al. (2001), simple large-scale background structures at early levels are referred to here as “segments”; following Rensink and Enns (1995), more complex localized structures are referred to as “proto-objects.” “Object” (as in “visual object”) is reserved for the highest level of visual structure, viz., a coherent representation in which all components refer—over both space and time—to a single underlying substrate, ideally capturing the structure of a physical object in the world.
2. The function-centered categorization developed here echoes to a large extent the approach advocated by Marr (1982) and Dennett (1994). This begins by analyzing the process in terms of its function, followed by analysis of the mechanisms/representations that carry it out, followed by analysis of the underlying neural implementation. Given the focus of this chapter, only the first two are discussed in the framework here.

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